HW06

知识点1:

1. 杂质(掺杂)电离本征激发，$np=n_i^2$(热平衡状态下的载流子浓度关系)

3. 正向，反向，反向

4. C. $v_2-v_1=2.3V_T\lg\frac{I_2}{I_1}$，$V_T=25mV$.

5. C.

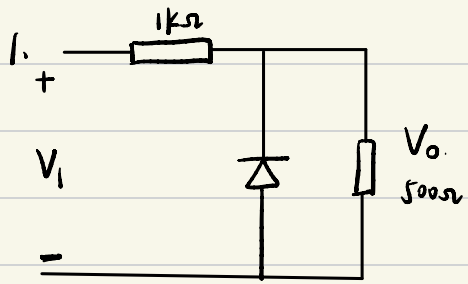
7. 0. $+\infty$

8. D

10. $r_d=\frac{nV_T}{I_D}$，$I_D\uparrow$ $r_d\downarrow$，B.

知识点2:

1.



(1). $V_1=10V$. $V_o=10V\times\frac{500}{500+1000}=3.3V$.

$V_1=15V$, $V_o=15V\times\frac{500}{500+1000}=5V$.

$V_1=35V$. $35\times\frac{500}{500+1000}=\frac{35}{3}V>6V$

则 $V_o=V_Z=6V$.

(2). $I=\frac{35V-6V}{1k\Omega}=29mA>25mA$

则二极管烧毁.

2. 半波，全波，桥式

5. 对于D1: $-3+1=-2<0.7$

对于D2: $-3+1=-2<0.7$

则 $V_{omax}=1V$, $V_{omin}=-1V$

6. $-10 < v_i < 0$, $v_o = 0$, $i_i = \frac{v_i}{10k\Omega} = 0.1 v_i$ mA.

$0 < v_i < 3$, $v_o = 0$, $i_i = 0$ mA

$3 < v_i < 10$, $v_o = \frac{v_i - 3}{2}$, $i_i = \frac{v_o}{10} = \frac{v_i - 3}{20}$ mA

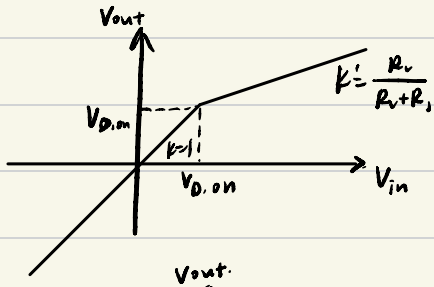
7. D

8. $I_D = \frac{10V - 0.7V}{10k\Omega} = 0.93$

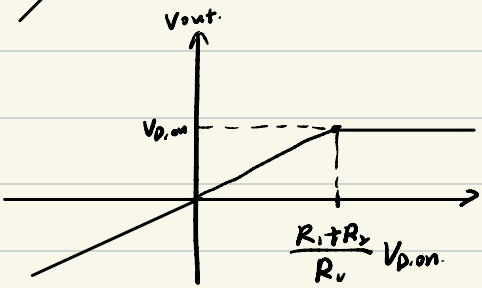
$r_d = \frac{nV_T}{I_D} = \frac{25}{0.93} = 26.9\Omega$.

$10V \times \frac{26.9\Omega}{26.9\Omega + 10k\Omega} = 0.0268$ V.

9.



10.



11. $I_D = \frac{5.7V - 0.7V}{1k\Omega} = 5mA$.

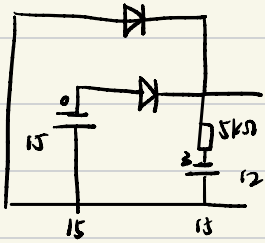
$r_d = \frac{nV_T}{I_D} = \frac{25mV}{5mA} = 5\Omega$.

$i_d = \frac{v_i}{r_d} = \frac{10mV}{5\Omega}$

$= 2mA$

12. C.

13. B.



14. $\frac{2\times10\times\sqrt{2}-2\times10}{2}=13.44V$

15. 整流器、滤波器

16. $D_1: 10I\cdot e^{\frac{V_{D2}-V}{V_T}}=2$

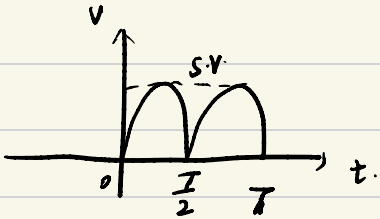
$D_2: I\cdot e^{\frac{V_{D2}}{V_T}}=8$

解得 $V=V_T\cdot\ln 40$

则 $10e^{\frac{V}{V_T}}=\frac{I_v}{10-I_v}$

解得 $I_v=5.75mA$

18. D.

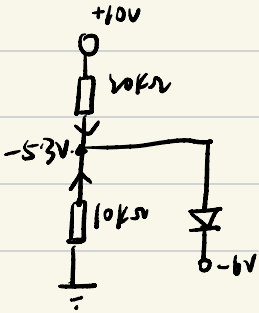


20. $I=\frac{600mV}{100\Omega}=6mA$

$r_d=\frac{nV_T}{I}=\frac{25mA}{6mA}=4.17\Omega$

$V=\frac{4.17\times3}{4.17\times3+100}\times100$

$=11.12mV$

21. A.

22. $V_A=-5.3V$. $I_D=1.3mA$